

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: AI-Powered Career Intelligence And Adapative Skill-Gap Analysis Platform For Students

Abstract

*The rapid growth of technology and the evolving demands of the job market have made it increasingly difficult for students to identify the skills required for their desired careers. Many students graduate without a clear understanding of industry expectations, resulting in skill gaps that reduce employability. The proposed project, **AI-Powered Career Intelligence and Adaptive Skill-Gap Analysis Platform for Students**, aims to address this challenge by providing personalized career guidance, skill-gap analysis, and learning recommendations using Artificial Intelligence (AI).*

The platform allows students to create a profile by entering their academic information, technical skills, certifications, projects, interests, and career aspirations. Using AI-based analysis, the system compares the student's current skill set with the requirements of various career roles such as Software Engineer, Data Analyst, AI Engineer, Cybersecurity Analyst, and Full Stack Developer. It identifies missing skills, evaluates the student's readiness level, and generates a personalized learning roadmap to help bridge the identified gaps.

The proposed system consists of modules including student registration and authentication, profile management, AI-powered career recommendation, adaptive skill-gap analysis, personalized learning roadmap generation, progress tracking, and an administrator dashboard.

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The expected outcome of this project is an intelligent career guidance platform that enables students to make informed career decisions, identify their strengths and weaknesses, and systematically improve their employability. By delivering personalized recommendations instead of generic advice, the platform helps students focus on acquiring industry-relevant skills and tracking their progress toward career readiness.

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References (Category specific)

- [1] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, USA, 2016, pp. 785–794.
- [2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [3] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
- [4] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ, USA: Pearson, 2021.
- [5] M. Lutz, *Learning Python*, 5th ed. Sebastopol, CA, USA: O'Reilly Media, 2013.
- [6] J. W. Lai, R. D. H. Gagero, L. Zhang, C. C. Sze, and F. S. Lim, "AI-Assisted Career Preparation and Skill Gap Awareness: A Retrospective Pretest-Posttest Study," *Education Sciences*, vol. 15, no. 2, Art. no. 191, 2025.

Project Team Members

S. No.	Regd. No.	Name of the Student	Signature
1	23BQ1A61D5	T. LEELA KUMAR REDDY	
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Project Guide

Name:- Mr. Syed Beeban Basha

Signature:-

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INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.